

MATH1014 Calculus II Tutorial

Week 12

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1 Review

1.1 Power Series and Interval of Convergence

A **power series** about $x = a$ is a series of the form:

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \dots$$

- **Radius of Convergence (R):** We typically use the Ratio Test or Root Test to find R . By solving $\lim_{n \rightarrow \infty} \left| \frac{c_{n+1}(x-a)^{n+1}}{c_n(x-a)^n} \right| < 1$, we get $|x-a| < R$.
- **Interval of Convergence:** The series converges absolutely for $x \in (a-R, a+R)$.
- **Endpoints:** The Ratio/Root tests are inconclusive at the boundaries $x = a-R$ and $x = a+R$. You **must** manually plug these values into the original series and use other tests (e.g., Alternating Series Test, p -series) to determine if they are included in the interval.

1.2 Taylor and Maclaurin Series

If $f(x)$ has derivatives of all orders at $x = a$, the **Taylor Series** of $f(x)$ centered at $x = a$ is:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots$$

When $a = 0$, it is called the **Maclaurin Series**. You should memorize these common Maclaurin Series:

- $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots \quad \text{for } |x| < 1$
- $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots \quad \text{for all } x$
- $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \quad \text{for } x \in (-1, 1]$

1.3 Operations on Power Series

Within its interval of convergence, a power series can be treated like a polynomial.

- **Substitution:** You can substitute x with $2x$, $-x^2$, etc., into a known series to find a new series.
- **Differentiation and Integration:** You can differentiate or integrate a power series term-by-term.

$$\frac{d}{dx} \left[\sum c_n (x-a)^n \right] = \sum n c_n (x-a)^{n-1}$$
$$\int \left[\sum c_n (x-a)^n \right] dx = C + \sum c_n \frac{(x-a)^{n+1}}{n+1}$$

Important: The radius of convergence R remains the same after differentiation or integration, but convergence at the endpoints might change!

2 Exercises

2.1 Finding Taylor Expansions and Intervals of Convergence

Exercise 2.1. Find the Taylor expansion of the following functions at the given point and determine the interval of convergence.

1. $\frac{x}{2-x-x^2}$ at $x=0$
2. $\frac{x-1}{x+1}$ at $x=1$
3. $(1+x)\ln(1-x)$ at $x=0$

Solution. (1) Idea. Factor the denominator, use partial fractions, and apply the geometric series expansion $\frac{1}{1-r} = \sum r^n$.

First, factor the denominator: $2-x-x^2 = (2+x)(1-x)$. Now, set up the partial fraction decomposition:

$$\frac{x}{(2+x)(1-x)} = \frac{A}{1-x} + \frac{B}{2+x} \implies x = A(2+x) + B(1-x)$$

Setting $x=1 \implies 1=3A \implies A=1/3$.

Setting $x=-2 \implies -2=3B \implies B=-2/3$. Thus, the function can be rewritten as:

$$f(x) = \frac{1}{3} \left(\frac{1}{1-x} \right) - \frac{2}{3} \left(\frac{1}{2+x} \right)$$

We expand each term into a Maclaurin series. For the first term:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{for } |x| < 1$$

For the second term, we must rewrite it to look like $\frac{1}{1-r}$:

$$\frac{1}{2+x} = \frac{1}{2(1+\frac{x}{2})} = \frac{1}{2} \cdot \frac{1}{1-(-\frac{x}{2})} = \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n \quad \text{for } \left|-\frac{x}{2}\right| < 1 \implies |x| < 2$$

Combining both series:

$$\begin{aligned} f(x) &= \frac{1}{3} \sum_{n=0}^{\infty} x^n - \frac{2}{3} \sum_{n=0}^{\infty} \frac{(-1)^n}{2^{n+1}} x^n \\ &= \frac{1}{3} \sum_{n=0}^{\infty} \left[1 - 2 \cdot \frac{(-1)^n}{2^{n+1}} \right] x^n \\ &= \frac{1}{3} \sum_{n=0}^{\infty} \left[1 + \frac{(-1)^{n+1}}{2^n} \right] x^n \end{aligned}$$

Interval of Convergence: The first series converges for $|x| < 1$ and the second for $|x| < 2$. The combined series converges on the intersection, which is $|x| < 1$, so $R = 1$. Now we check the endpoints $x = 1$ and $x = -1$:

- If $x = 1$: The series becomes $\frac{1}{3} \sum_{n=0}^{\infty} \left[1 + \frac{(-1)^{n+1}}{2^n} \right] (1)^n$. As $n \rightarrow \infty$, the n -th term approaches $1 \neq 0$. By the Test for Divergence, it diverges.
- If $x = -1$: The series becomes $\frac{1}{3} \sum_{n=0}^{\infty} \left[1 + \frac{(-1)^{n+1}}{2^n} \right] (-1)^n$. As $n \rightarrow \infty$, the term does not approach 0 (it oscillates near ± 1). It diverges.

Thus, the interval of convergence is $(-1, 1)$.

(2) Idea. We need the expansion at $x = 1$, which means powers of $(x - 1)$. We algebraically manipulate the function to isolate $(x - 1)$.

$$f(x) = \frac{x-1}{x+1} = \frac{x-1}{(x-1)+2} = \frac{x-1}{2} \cdot \frac{1}{1 + \frac{x-1}{2}}$$

Now we use the geometric series formula $\frac{1}{1-r} = \sum r^n$ with $r = -\frac{x-1}{2}$:

$$\begin{aligned} f(x) &= \frac{x-1}{2} \sum_{n=0}^{\infty} \left(-\frac{x-1}{2} \right)^n \\ &= \frac{x-1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^n}{2^n} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (x-1)^{n+1}}{2^{n+1}} \end{aligned}$$

To make the sum look like standard powers of n , let $k = n + 1$. When $n = 0$, $k = 1$:

$$f(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} (x-1)^k}{2^k}$$

Interval of Convergence: The geometric series converges when $|r| < 1 \implies \left| -\frac{x-1}{2} \right| < 1 \implies |x-1| < 2$. Thus $R = 2$. The open interval is $(-1, 3)$. Now check the endpoints:

- If $x = 3$: Series is $\sum_{k=1}^{\infty} \frac{(-1)^{k-1} (2)^k}{2^k} = \sum_{k=1}^{\infty} (-1)^{k-1}$. The limit of the terms is not 0, so it diverges.
- If $x = -1$: Series is $\sum_{k=1}^{\infty} \frac{(-1)^{k-1} (-2)^k}{2^k} = \sum_{k=1}^{\infty} (-1)^{k-1} (-1)^k = \sum_{k=1}^{\infty} (-1)^{2k-1} = \sum_{k=1}^{\infty} -1$. It diverges.

Thus, the interval of convergence is $(-1, 3)$.

(3) Idea. Use the known Maclaurin series for $\ln(1 - x)$ and multiply it by $(1 + x)$.

Recall the standard series:

$$\ln(1 + x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$$

Substituting x with $-x$:

$$\ln(1 - x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} (-x)^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} (-1)^n x^n}{n} = \sum_{n=1}^{\infty} -\frac{x^n}{n}$$

Now multiply by $(1 + x)$:

$$\begin{aligned} (1 + x) \ln(1 - x) &= (1 + x) \sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) x^n \\ &= \sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) x^n + x \sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) x^n \\ &= \sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) x^n + \sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) x^{n+1} \end{aligned}$$

To combine them, shift the index of the second sum. Let $k = n + 1 \implies n = k - 1$. When $n = 1, k = 2$:

$$\sum_{n=1}^{\infty} \left(-\frac{1}{n}\right) x^{n+1} = \sum_{k=2}^{\infty} \left(-\frac{1}{k-1}\right) x^k$$

Now pull out the $n = 1$ term from the first sum so both sums start at 2:

$$\begin{aligned} (1 + x) \ln(1 - x) &= -x + \sum_{n=2}^{\infty} \left(-\frac{1}{n}\right) x^n + \sum_{n=2}^{\infty} \left(-\frac{1}{n-1}\right) x^n \\ &= -x - \sum_{n=2}^{\infty} \left(\frac{1}{n} + \frac{1}{n-1}\right) x^n \end{aligned}$$

Interval of Convergence: The standard series for $\ln(1 - x)$ has $R = 1$, so the open interval is $(-1, 1)$. Check the endpoints:

- If $x = 1$: The series evaluates to $-1 - \sum_{n=2}^{\infty} \left(\frac{1}{n} + \frac{1}{n-1}\right)$. Since $\sum \frac{1}{n}$ is a divergent harmonic series, this sum diverges. (Also, $\ln(1 - x)$ is undefined at $x = 1$).
- If $x = -1$: The series evaluates to $1 - \sum_{n=2}^{\infty} \left(\frac{1}{n} + \frac{1}{n-1}\right) (-1)^n$. This is an alternating series. The terms $b_n = \frac{1}{n} + \frac{1}{n-1}$ are decreasing and limit to 0. By the AST, it converges.

Thus, the interval of convergence is $[-1, 1)$. □

2.2 Evaluating Series Sums using Taylor Series Operations

Exercise 2.2. Use the Taylor expansion of $\frac{e^x - 1}{x}$ at $x = 0$ to show $\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = 1$.

Solution. Idea. Start with the Maclaurin series for e^x , manipulate it algebraically to form the function, differentiate it to generate the factor n , and plug in a specific value for x .

We know the Maclaurin series for e^x :

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Subtract 1 and divide by x :

$$\begin{aligned} \frac{e^x - 1}{x} &= \frac{1}{x} \left(\sum_{n=0}^{\infty} \frac{x^n}{n!} - 1 \right) \\ &= \frac{1}{x} \left(\sum_{n=1}^{\infty} \frac{x^n}{n!} \right) \\ &= \sum_{n=1}^{\infty} \frac{x^{n-1}}{n!} \end{aligned}$$

To match the denominator $(n+1)!$ in the target expression, let's shift the index. Let $k = n - 1 \implies n = k + 1$. When $n = 1$, $k = 0$:

$$\frac{e^x - 1}{x} = \sum_{k=0}^{\infty} \frac{x^k}{(k+1)!}$$

We want to generate an n (or k) in the numerator. We can do this by taking the derivative of both sides with respect to x . On the right side (differentiating term-by-term):

$$\frac{d}{dx} \left[\sum_{k=0}^{\infty} \frac{x^k}{(k+1)!} \right] = \sum_{k=1}^{\infty} \frac{kx^{k-1}}{(k+1)!}$$

On the left side (using the Quotient Rule):

$$\frac{d}{dx} \left(\frac{e^x - 1}{x} \right) = \frac{e^x(x) - (e^x - 1)(1)}{x^2} = \frac{xe^x - e^x + 1}{x^2}$$

Equating both sides:

$$\frac{xe^x - e^x + 1}{x^2} = \sum_{k=1}^{\infty} \frac{kx^{k-1}}{(k+1)!}$$

To evaluate the series $\sum_{k=1}^{\infty} \frac{k}{(k+1)!}$, we simply plug in $x = 1$:

$$\sum_{k=1}^{\infty} \frac{k(1)^{k-1}}{(k+1)!} = \frac{(1)e^1 - e^1 + 1}{1^2} = \frac{e - e + 1}{1} = 1$$

Thus, $\sum_{n=1}^{\infty} \frac{n}{(n+1)!} = 1$. □

Exercise 2.3. Evaluate the following series.

1. $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{2^n}$
2. $\sum_{n=2}^{\infty} (-1)^n \frac{1}{2^n(n^2 - 1)}$

Solution. (1) Idea. Construct a function $f(x)$ whose series expansion matches the given sum when $x = 1/2$. We will use integration to remove the n in the numerator, identify the resulting geometric series, and then differentiate back to find $f(x)$.

Let $f(x) = \sum_{n=1}^{\infty} (-1)^{n-1} n x^n$. Our goal is to find $f(1/2)$. To eliminate the n , we divide by x to get x^{n-1} , which is the exact derivative of x^n . Let $g(x) = \frac{f(x)}{x} = \sum_{n=1}^{\infty} (-1)^{n-1} n x^{n-1}$. Now, integrate $g(x)$ term-by-term:

$$\begin{aligned}\int g(x) dx &= \int \left(\sum_{n=1}^{\infty} (-1)^{n-1} n x^{n-1} \right) dx \\&= \sum_{n=1}^{\infty} (-1)^{n-1} \int n x^{n-1} dx \\&= \sum_{n=1}^{\infty} (-1)^{n-1} x^n + C \\&= x - x^2 + x^3 - x^4 + \cdots + C\end{aligned}$$

This is a geometric series with first term $a = x$ and common ratio $r = -x$.

$$\int g(x) dx = \frac{x}{1 - (-x)} + C = \frac{x}{1 + x} + C$$

To find $g(x)$, we differentiate both sides:

$$g(x) = \frac{d}{dx} \left(\frac{x}{1+x} + C \right) = \frac{1(1+x) - x(1)}{(1+x)^2} = \frac{1}{(1+x)^2}$$

Since $g(x) = f(x)/x$, we have:

$$f(x) = x \cdot g(x) = \frac{x}{(1+x)^2}$$

Finally, substitute $x = 1/2$ into $f(x)$ to get the sum of the series:

$$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{n}{2^n} = f\left(\frac{1}{2}\right) = \frac{1/2}{(1+1/2)^2} = \frac{1/2}{9/4} = \frac{2}{9}.$$

(2) Idea. Construct $f(x) = \sum_{n=2}^{\infty} (-1)^n \frac{1}{n^2-1} x^n$, so the answer is $f(1/2)$. We will multiply by x , differentiate to remove $(n+1)$ from the denominator, and then integrate to reconstruct the function.

Note that $n^2 - 1 = (n-1)(n+1)$. Let $f(x) = \sum_{n=2}^{\infty} (-1)^n \frac{1}{(n-1)(n+1)} x^n$. Multiply by x to match the power $n+1$ with the denominator:

$$g(x) = x f(x) = \sum_{n=2}^{\infty} (-1)^n \frac{1}{(n-1)(n+1)} x^{n+1}$$

Differentiate $g(x)$:

$$\begin{aligned}g'(x) &= \sum_{n=2}^{\infty} (-1)^n \frac{n+1}{(n-1)(n+1)} x^n = \sum_{n=2}^{\infty} (-1)^n \frac{1}{n-1} x^n \\&= \sum_{n=2}^{\infty} (-1)^n \frac{1}{n-1} x \cdot x^{n-1} \\&= x \sum_{n=2}^{\infty} (-1)^n \frac{x^{n-1}}{n-1}\end{aligned}$$

Let $k = n - 1$. When $n = 2, k = 1$:

$$g'(x) = x \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k}$$

Recall the Maclaurin series for $\ln(1+x) = \sum_{k=1}^{\infty} (-1)^{k+1} \frac{x^k}{k}$. Since $(-1)^{k+1} = (-1)^{k-1}$, we have:

$$g'(x) = x \ln(1+x)$$

Now, integrate $g'(x)$ to find $g(x)$. Use integration by parts: let $u = \ln(1+x), dv = x dx \implies du = \frac{1}{1+x} dx, v = \frac{x^2}{2}$.

$$\begin{aligned} g(x) &= \int x \ln(1+x) dx = \frac{x^2}{2} \ln(1+x) - \frac{1}{2} \int \frac{x^2}{1+x} dx \\ &= \frac{x^2}{2} \ln(1+x) - \frac{1}{2} \int \frac{x^2 - 1 + 1}{1+x} dx \\ &= \frac{x^2}{2} \ln(1+x) - \frac{1}{2} \int \left(x - 1 + \frac{1}{1+x} \right) dx \\ &= \frac{x^2}{2} \ln(1+x) - \frac{1}{2} \left(\frac{x^2}{2} - x + \ln(1+x) \right) + C \\ &= \frac{1}{2} (x^2 - 1) \ln(1+x) - \frac{1}{4} x^2 + \frac{x}{2} + C \end{aligned}$$

Since $g(x) = x f(x)$ has its lowest degree term starting at x^3 , $g(0) = 0$. Evaluating at $x = 0$: $g(0) = \frac{1}{2}(-1) \ln(1) - 0 + 0 + C = C \implies C = 0$. So $g(x) = \frac{1}{2}(x^2 - 1) \ln(1+x) - \frac{1}{4}x^2 + \frac{x}{2}$. Since $f(x) = \frac{g(x)}{x}$:

$$f(x) = \frac{x^2 - 1}{2x} \ln(1+x) - \frac{1}{4}x + \frac{1}{2}$$

Finally, evaluate at $x = 1/2$:

$$\begin{aligned} f(1/2) &= \frac{(1/4) - 1}{1} \ln(3/2) - \frac{1}{8} + \frac{1}{2} \\ &= -\frac{3}{4} \ln(3/2) + \frac{3}{8} = \frac{3}{8} - \frac{3}{4} \ln\left(\frac{3}{2}\right) \end{aligned}$$

□

Exercise 2.4. Let $a_n = c + (n-1)d, n = 1, 2, \dots$, and $b > 1$. Find the value of $\sum_{n=1}^{\infty} \frac{a_n}{b^n}$.

Solution. Idea. Split the sum into two parts. The first is a standard geometric series. The second requires differentiating a geometric series to generate the $(n-1)$ term.

$$\sum_{n=1}^{\infty} \frac{c + (n-1)d}{b^n} = c \sum_{n=1}^{\infty} \frac{1}{b^n} + d \sum_{n=1}^{\infty} \frac{n-1}{b^n} = I + II$$

For I , it is a geometric series with first term $a = 1/b$ and common ratio $r = 1/b$. Since $b > 1$, $|r| < 1$.

$$I = c \cdot \frac{1/b}{1 - 1/b} = c \cdot \frac{1/b}{(b-1)/b} = \frac{c}{b-1}$$

For II , the $n = 1$ term is 0, so we can start the sum from $n = 2$:

$$II = d \sum_{n=2}^{\infty} \frac{n-1}{b^n}$$

Let's define a function $f(x) = \sum_{n=2}^{\infty} \frac{n-1}{b^n} x^{n-2}$, so that $II = d \cdot f(1)$. We integrate $f(x)$ to remove the $(n-1)$ multiplier:

$$\begin{aligned} \int_0^x f(t) dt &= \sum_{n=2}^{\infty} \frac{n-1}{b^n} \int_0^x t^{n-2} dt = \sum_{n=2}^{\infty} \frac{n-1}{b^n} \left[\frac{t^{n-1}}{n-1} \right]_0^x \\ &= \sum_{n=2}^{\infty} \frac{x^{n-1}}{b^n} = \frac{1}{b} \sum_{n=2}^{\infty} \left(\frac{x}{b} \right)^{n-1} \end{aligned}$$

This is a geometric series with first term $a = x/b^2$ (when $n = 2$) and common ratio $r = x/b$.

$$\int_0^x f(t) dt = \frac{1}{b} \cdot \frac{x/b}{1 - x/b} = \frac{1}{b} \cdot \frac{x/b}{(b-x)/b} = \frac{x}{b(b-x)}$$

To find $f(x)$, we differentiate:

$$f(x) = \frac{d}{dx} \left[\frac{x}{b(b-x)} \right] = \frac{1}{b} \frac{1(b-x) - x(-1)}{(b-x)^2} = \frac{1}{b} \frac{b}{(b-x)^2} = \frac{1}{(b-x)^2}$$

Thus, $II = d \cdot f(1) = \frac{d}{(b-1)^2}$. Combining both parts:

$$\sum_{n=1}^{\infty} \frac{a_n}{b^n} = I + II = \frac{c}{b-1} + \frac{d}{(b-1)^2} = \frac{c(b-1) + d}{(b-1)^2}.$$

□